Interpretable Multimodal Machine Learning Framework for Cognitive Severity Estimation in Alzheimer's Disease Using MRI-Derived Brain Features and Blood Biomarkers

Sarvesh Kumar*, Adarsh Anand†, Dr. Niranjana G‡
*Department of Computing Technologies, SRM Institute of Science and Technology, Chennai, India
Email: sk6476@srmist.edu.in
†Department of Computing Technologies, SRM Institute of Science and Technology, Chennai, India
Email: aa0759@srmist.edu.in
‡Department of Computing Technologies, SRM Institute of Science and Technology, Chennai, India
Email: niranjag@srmist.edu.in

***Abstract*—Alzheimer disease (AD) is a progressive neurodegenerative disease that is associated with memory impairment, cognitive impairment and atrophy of the brain in its structure. Conventional clinical assessment methods do not allow early diagnosis. This paper presents a multimodal machine learning model that would combine both structural MRI images and blood biomarkers early diagnosis of Alzheimer disease. The suggested model involves modality-sensitive preprocessing, features selection based on SHAP-based importance ranking, and classification based on ensemble models, including XGBoost and Random Forest. Predictive accuracy The experimental test shows that multimodal fusion is highly superior to the unimodal models. The framework proposes an interpretable, scaled, and non-invasive framework of AI-aided neurodiagnostic. Index Terms—Alzheimer's Disease, MRI, Blood Biomarkers, Machine Learning, Multimodal Learning, XGBoost, Explainable AI, ROC-AUC.**

***Index Terms*—Alzheimer's Disease, MRI, Blood Biomarkers, Machine Learning, Multimodal Learning, XGBoost, Explainable AI, ROC-AUC**

I. Introduction

Alzheimer disease (AD) is the major cause of dementia it is said to be a global health hazard that is rapidly increasing, especially in the aging society. AD has a clinical presentation of progressive impairment of memory, impairment of executive functions, language disorders, and, eventually, capacity to live independently. On the biological level, the disease is distinguished by massive neuronal loss and synaptic degeneration with formation of substantial brain structural and functional changes. Neuropathological features of AD are extracellular deposition of amyloid-beta (A 2) plaques and intracellular deposits of neurofibrillary tangles consisting of hyperphosphorylated tau protein. These pathophysiological alterations disorganize the communication between the neurons, dysfunction axonal transport, elicit neuroinflammatory responses with an irreversible neuronal death.

The most significantly affected structures are the structures of the hippocampus and the medial temporal lobe which are the core of memory consolidation and spatial navigation. With further degeneration, there is atrophy of cortical parts, such as the temporal, parietal, and frontal lobes which can be measured. This type of structural degradation can be observed with magnetic resonance imaging (MRI), especially with help of volumetric assessment of the hippocampal shrinkage and cortical thinning. These quantitative changes in the anatomy are one of the main bases of our project, and structural characteristics determined by MRI are one of the main modalities to identify and classify diseases.

Traditional diagnostic methods are still limited in spite of improvements that have been made on the neuroimaging and biomarker discovery. Neuropsychological examinations like the Mini-Mental State Examination (MMSE) and other cognitive batteries are more commonly applied although they may be detected too late when the neurons are already significantly damaged. Positron emission tomography (PET) examination offers visualization of the amyloid and tau deposit, but is expensive, and not always available. Cerebrospinal fluid (CSF) tests provide a biochemical validation of the presence of pathological changes and due to its invasive lumbar puncture, it is not the most promising choice in the context of mass screening. All these restrictiveness illustrate that there is a serious clinical gap that needs to be filled: means of precise, scalable and the least invasive early detection.

We use computer-aided multimodal framework of machine learning through a response to this issue to incorporate structural MRI characteristics alongside blood-based biomarkers in our framework. The use of this integration is based on the fact that Alzheimer is a multifactorial disease. The macroscopic anatomical changes that Structural MRI can capture include atrophy of the hippocampal, enlargement of ventricles, and cortical thinning. Conversely, blood biomarkers indicate systemic biochemical changes with an example being amyloid-beta ratios, tau protein levels, inflammatory markers, and neurodegeneration-associated proteins. Each of these

modalities is partially informative about disease pathology, but through their combination, they can be used to provide complementary information which would increase predictive power.

The set of suggested system has preprocessing pipelines that are modality-specific. The data provided with MRI are normalized and feature extracted (e.g., volumetric measures and regional measures) and dimensionality reduced to guarantee strong representation of structural patterns of degeneration. At the same time, the data of blood biomarkers is purified with imputation of missing values, exclusion of outliers and standardization to ensure the integrity of the statistics. Each of the modalities is processed individually with the feature selection methods such as correlation and SHAP-based importance ranking to remove redundancy and show clinically significant predictors.

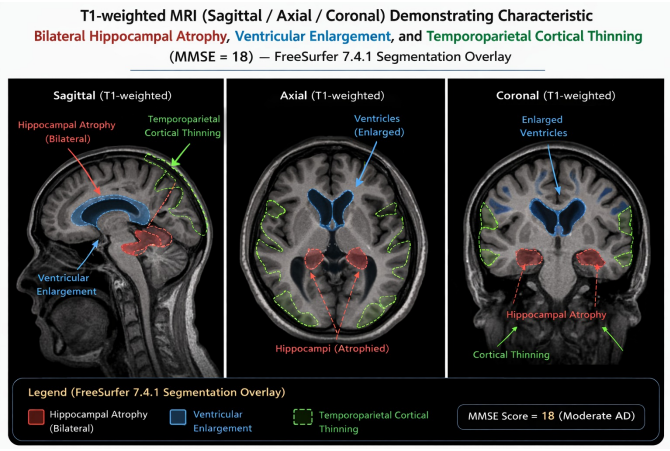

Fig. 1. T1-weighted MRI views (Sagittal, Axial, and Coronal) showing structural markers of Alzheimer's disease, including bilateral hippocampal atrophy, ventricular enlargement, and temporoparietal cortical thinning obtained using FreeSurfer 7.4.1 segmentation (MMSE = 18).

The modality-specific features are then used to train machine learning models, especially the ensemble models, including Random Forest and XGBoost. These models have been selected due to their capability to deal with nonlinear relations, high-dimensional data and imbalance of classestypical biomedical datasets challenges. Multimodal fusion strategy combines output of the MRI based and biomarker based classifiers to form a final prediction that either classifies subjects as either Alzheimer disease, mild cognitive impairment (MCI) or as cognitively normal (CN). Through probabilistic outputs or an addition of optimized features, the system integrates diagnostic complementary signals in both structural and biochemical approaches.

Notably, the interpretability is implemented into the framework with the SHAP (SHapley Additive exPlanations) analysis that allows determining the most powerful and important MRI regions and biomarker variables that increase model predictions. This is an essential step toward clinical acceptance, that is, because it will put the computational results in conformity with an established body of neuropathological evidence the central role of hippocampal volume reduction and high tau in disease progression.

In such a way, this project explicitly focuses on the shortcomings of the classic diagnostic pathways by coming up with explainable, multimodal, and non-invasive AI framework that is data-driven. Through techniques of combining structural neuroimaging and peripheral biomarkers into a single machine learning system, it should be possible to better detect the disease at its initial stages and justify its scalable screening measures in the diagnosis of Alzheimer disease.

II. Literature Review

A. Machine Learning with Structural MRI in Alzheimer's Detection

Initial computational research into the disease of Alzheimer focused on structural magnetic resonance imaging (MRI) as the main diagnostic modality because of its ability to provide more detailed anatomical changes in response to neurodegeneration. MRI allows quantification of hippocampal atrophy, cortical atrophy, ventricular hypertrophy, and reduction of the gray matter which are well-defined structural markers of Alzheimer pathology. Early researchers have isolated handcrafted features that included regional brain volumes, and cortical thickness and made use of these as inputs to advertise learning algorithms with antique machine knowledge, like Support Vector Machines (SVM), Logistic Regression and decision tree learning classifiers. These methods showed structural patterns obtained with the help of MRI could be translated into predictive models that could be used to differentiate between Alzheimer and cognitively normal individuals. However, these initial models had a number of restrictions, such as small summary size, reduced statistical power, and enhanced method to overfitting. Manual feature engineering had been very feature-specific and dataset-specific improving only reproducibility and not cross relationship across institutions. Also, the correct diagnosis of Mild Cognitive Impairment (MCI) was difficult since at this point structural changes are tiny and diffuse. As a result, MRI-single models had lower sensitivity in detecting cancer at the earliest stage, which is why more sophisticated computational methods, as well as the incorporation of additional biological evidence, should be developed.

B. Deep Learning Advancements in Neuroimaging

Deep learning created the first major advancement in neuroimaging-based detection of Alzheimer since it allows the automatic learning of features starting with raw MRI data. Convolutional Neural Networks (CNNs), specifically three dimensional ones, have been implemented to handle volumetric brain scans in order to enable the hierarchical spatial representations of structural degeneration to be learnt without depending on the handcrafted features. These models enhanced the classification performance particularly when separating advanced stages of Alzheimer disease and normal controls, which were characterized by identification of intricate fashion of atrophy that are stipulated in multiple areas of the brain.

Transfer learning and data augmentation techniques were used to overcome the lack of data availed, which increased the resilience of models. Although the accuracy of predictions has been improved, deep learning models pose some essential challenges in the field of clinical scenarios. Internal decision processes tend to be opaque that gives little explanation of what anatomy leads to predictions. This does not enhance transparency and it constrained practical implementation of healthcare setups. Also, deep neural networks consume massive resources in terms of computational power as well as huge datasets with annotated data to learn to generalize. Such limitations indicate the necessity of a compromise between predictive performance and interpretability and computability. Our project in turn combines both ensemble learning and SHAP-based explainability to yield both high performance in classification and clinical insights on the structural factors of Alzheimer disease.

C. Peripheral Blood Biomarkers as Predictive Indicators

In addition to the neuroimaging, studies have turned to peripheral blood biomarkers as the minimally invasive predictors of the Alzheimer pathology. With modern improvements in technology in the assay sensitivity, it is possible to measure plasma amyloid-beta ratios, phosphorylated tau isoforms, total tau concentrations and neurofilament light chain concentrations reliably. These biomarkers indicate the underlying pathological processes including the aggregation of proteins, damage to axons and neuroinflammatory processes related to disease progression. Blood-related tests have a great benefit over cerebrospinal fluid where they are less invasive, lower in prices and more applicable in screening a large population. There are machine learning models on biochemical datasets which have been found to have high sensitivity when it comes to identifying risk persons of Alzheimer disease. The drawbacks of biomarker-only methods are because of variability conditions caused by comorbid chronic ailments, demographics, and systemic inflammation, decreasing specificity. Notably, biochemical markers do not reflect anatomical alterations in the brain directly and have no spatial resolution that can be used to determine structural degeneration. Individual biomarker models can therefore be potentially incapable of differentiating early neurodegenerative changes and normal aging or other extrasystemic pathologies. Such restrictions accentuate the need to combine blood biomarkers along with the structural MRI characteristics in order to establish a more substantial and biologically holistic-based predictive paradigm.

D. Multimodal Integration Strategies

Since Alzheimer disease is a multifactorial disease, in recent years multimodal data integration has been given greater consideration to reflect the complementary evident components of neurodegeneration. Using the structural imaging characteristic with the biochemical signs, multimodal systems try to demonstrate both the anatomy and the molecular size of the disease development. Early fusion and late fusion are two major fusion strategies that are commonly adopted. Early fusion assembles modalities features together before the training of a model, allowing the algorithm to be trained which may learn joint feature representations. Late fusion sums or averages the prediction of modality-specific models, and consequently modality independence is maintained, although at the cost of improved robustness. It is empirically indicated that multimodal techniques are more accurate and sensitive systems than unimodal systems to aid in classification and detecting the item at early stages. Yet, there are still methodological inconsistencies found in all studies, such as poor modality-specific normalization and the lack of validation procedures. Under scaling of heterogeneous data types can be a source of bias and a decrease in the reliability. Moreover, there are multimodal frameworks that focus on predictive metrics and never combine explainability mechanisms. These vulnerabilities demonstrate that there has to be a close-knit design of architectures that are carefully cross-validated with the intuitive results. Our project will deal with these factors by deploying modality-directed preprocessing, modelled fusion and envelope-based modelling in a transparent and statistically validized framework.

E. Explainable Artificial Intelligence in Neurodegenerative Research

With AI being more deeply integrated into healthcare systems, interpretability has come out as a core condition to clinical adoption. Explainable Artificial Intelligence (XAI) attempts are designed to seek explanations on why a model will make a specific prediction and can help build the confidence of a physician and the regulation necessary to follow them as a trend. One of these techniques, SHAP (SHapley Additive Explanations), has become popular because of its theoretical roots in cooperation game theory and of its characteristic of measurably adding contributions of features to individual predictions. Explainability analyses have proven useful also in scientists of Alzheimer research by identifying brain atrophy of the hippocampal, as well as cortical thinning and tau levels, as significant predictors. Nevertheless, most of the studies that use explainable methods only deal with unimodal imaging models, which does not allow them to gain extensive knowledge on multimodal interactions. Also the interpretability is commonly regarded as an after-thought analysis as opposed to an inseparable component of model construction. Vigorous validation methods such as cross-validation and external testing have not been as emphatic which diminishes confidence in model generalization. Our model makes explainability integral to the modeling process, where SHAP analysis is used on ensemble models to clarify that both anatomical and biochemical features regulate the classification results.

F. Research Gap and Contribution

The critical review of the literature demonstrates that there are still gaps in the research on multimodal detection of Alzheimer. Although both, MRI-based and biomarker-based models display diagnostic potential, integrative systems between the two modes with detailed validation are relatively

scarce. Predictive accuracy is often given a high priority by the deep learning methodology thus limiting clinical translation. Also, poor use of cross-validation methods and data asymmetry can compromise the model generalizability. The current paper attempts to tackle these issues by suggesting a hierarchical multimodal framework, incorporating the structural features obtained using MRI along with the data on the blood-based biomarkers into a hierarchical machine learning structure. The preprocessing is modality-specific and provides suitable normalization and statistical integrity, whereas the feature selection decreases the redundancy and improves robustness. XGBoost and Random Forest are the ensemble techniques, which are used to model nonlinear interactions and heterogeneous distribution of features. Notably, SHAP-based analysis is included to offer insights on relative importance of predictors that are structural and biochemical that can be interpreted. By participating in the merging of explanatory data sources, systematic validation, and explainable modeling technologies, this research proposes to progress the creation of valid, universal, and clinically significant instruments of early Alzheimer disease.

III. Problem Statement

Assessment of cognitive impairment caused by Alzheimer disease at its early stages is a major clinical issue notwithstanding numerous innovations in the field of structural MRI analysis and biomarkers of blood examination. Existing methods of diagnosis like positron emission tomography (PET) scan and cerebral fluid tests are invasive, expensive, and can not be used in screening a large population. Current, unicomponent machine learning frameworks that only use MRI-derived structural or peripheral measures of biomarkers do not provide the intricate integration of anatomical neurodegeneration and molecular pathology biological underpinnings that contribute to the progression of Alzheimer disease. Consequently, these methods prove to be weak in describing Mild Cognitive impairment (MCI) which is the intermediate before clinical dementia. Moreover, the inability to interpret numerous predictive models would decrease the level of clinical trust and limit their use into the real-life healthcare setting. Hence, it is urgently required to have a scalable, explainable, as well as clinically interpretable multimodal machine learning framework that incorporates MRI structural data with blood-based biomarkers. This ought to support the ability to estimate cognitive severity accurately, the ability to evaluate disease at an early stage, and give biologically meaningful information to run in line with the known neuropathological pathways and thus ensure that there is the ability to monitor the Alzheimer disease at an accessible and deployable stage without involving any kind of invasive diagnostic protocols.

IV. Methodology

The suggested framework builds a decipherable multimodal machine learning framework to predict cognitive severity of Alzheimer disease through a combination of MRI-based structural stage brain characteristics and peripheral blood biomarkers. The research methodology is divided into several phases such as the data acquisition phase, preprocessing, feature engineering, multimodal integration, model training, evaluation, and interpretability analysis. From one stage to another, the anatomical and biochemical signs of neurodegeneration are both used to sound labor to predict the cognitive severity.

A. Data Acquisition

The initial step will be the collection of multimodal data i.e. structural MRI-based brain features and blood biomarkers measurements related to Alzheimer disease. Features derived through MRI reflect anatomical variations like volume of the hippocampal, cortical thickness, enlarged ventricles and density of the gray matter, which are generally accepted parameters of neurodegeneration. Blood biomarkers are values in terms of amyloid-betas ratios, tau protein ratio, and other biochemical values that can indicate neuronal damage and fairly inflammation. Such datasets are derived out of any publicly available Alzheimer research repositories or standard curated clinical datasets that have the relevant cognitive severity scores like the Mini-Mental State Examination (MMSE). The data samples contain both MRI-results of quantitative brain parameters and the values of blood biomarkers and the resultant cognitive severity score, which is used as the predictor variable.

B. Data Preprocessing

Preprocessing of the data is carried out prior to model training to guarantee the quality and consistency of data, both MRI-derived and biomarker. Such statistical imputation techniques as mean or median substitution are used to deal with missing values. This is done by statistically removing any outliers that might cause distortion of model training. Because the features and biomarker measurements have different numerical scales, the feature normalization and standardization methods including the z-score normalization is implemented. This is necessary to assure that all the variables will play their roles proportionately towards the learning process and features with higher numerical differences are not overridden.

C. MRI Feature Extraction

The Processing of structural MRI scans is done to retrieve some quantitative measurement of the brain, pertaining to the progression of the Alzheimer disease. Techniques of feature extraction calculate region-specific measures, such as volume of hippocampal and cortical thickness and ventricle size and gray matter density. These structural brain degeneration signs can be seen in these anatomical features because of cognitive breakdown. MRI features extracted are then turned into ordered numerical values which directly become inputs to machine learning algorithms.

D. Blood Biomarker Processing

The data of blood biomarkers is biochemical indicators of molecular processes in the pathology of Alzheimer. The data comprise biomarkers like amyloid-beta, tau protein and

neurodegeneration-associated biomarkers. This is so that these variables are standardized to have consistency numerically and to be comparable to MRI-derived features. The scaling of features is used in order to balance out the differences in data distribution between modalities.

E. Feature Selection

In order to enhance the effectiveness of models as well as reduce predictor redundancy, the practice of feature selection is used to determine the most applicable predictors. Redundant or strongly uninfluential variables are eliminated by application of statistical correlation analysis and feature importance methods. This process will decrease dimensions and optimize the performance of the models, because it involves the most informative structural and biochemical indicators.

F. Multimodal Feature Integration

Biomarker data and MRI-derived features are then may be combined after the step of preprocessing and feature selection that results in a single multimodal dataset. This integration facilitates the ability of the model to learn structural and biochemical patterns that are related to cognitive severity simultaneously. Multimodal feature fusion enables the machine learning model to indicate one-sided relationships between structural brain changes and molecular biomarkers.

Feature Fusion Representation: Denote the MRI feature vector by the symbol, X^{MRI} and the blood biomarker feature vector by, X^{Bio}. The multimodal feature representation is an integrated expression, which can be stated as:

$$X^{Multi} = [X^{MRI}, X^{Bio}] \tag{1}$$

Where:

- X^{MRI} = feature vector extracted from MRI imaging data
- X^{Bio} = feature vector obtained from blood biomarker measurements
- X^{Multi} = combined multimodal feature vector used as model input

The final multimodal feature matrix for the dataset can be expressed as:

$$X = \{X_1^{Multi}, X_2^{Multi}, ..., X_n^{Multi}\} \tag{2}$$

Where:

- n = total number of samples
- X_i^{Multi} = multimodal feature vector for the ith subject

Combining imaging-based structural features and biochemical biomarker measurements, the resulting multimodal data will be much more expressive of disease-associated patterns allowing machine learning models to gain better predictive advantages in estimating cognitive severity.

G. Model Training

Models of ensemble machine learning which utilize regression are used to predict cognitive severity scores. Algorithms like the XGBoost and the random forest are applied because they can use nonlinear relationships and high-dimensional data and non-uniform distributions of features. These ensemble methods are used in combination with predictions of several decision trees thereby enhancing prediction accuracy and lowing overfitting.

The data has been separated into training and testing applications (usually 80 training and 20 testing) so that they can be able to evaluate performance. The models acquire multimodal features which are complicated interactions between neuroimaging features, demographic and clinical indicators and the target variable which is the cognitive severity during training phase.

1. Random Forest Regression: Random Forest is an ensemble approach to learning, which builds several decision trees as part of the training process, and then arrives at the final prediction by averaging the decision of each tree.

$$\hat{y} = \frac{1}{T}\sum_{t=1}^{T} f_t(x) \tag{3}$$

Where:

- T = total number of decision trees in the forest
- $f_t(x)$ = prediction from the tth decision tree
- $\hat{y}$ = final predicted cognitive severity score
- x = input feature vector

Such an averaging process assists in lessening variance and enhances the generalization ability of the model.

2. XGBoost Regression: XGBoost happens to be based on gradient boosting algorithm and is structured in such a way that the trees are generated one after another with the subsequent tree trying to rectify the mistakes made by the earlier trees. The ultimate forecast is derived by adding the output of all the trees.

$$\hat{y}_i = \sum_{k=1}^{K} f_k(x_i) \tag{4}$$

Where:

- K = number of trees
- $f_k(x_i)$ = prediction of the kth tree for input x_i
- $\hat{y}_i$ = predicted cognitive score for sample i

The goal that is going to be optimized when training is:

$$Obj = \sum_{i=1}^{n} l(y_i, \hat{y}_i) + \sum_{k=1}^{K} \Omega(f_k) \tag{5}$$

Where:

- $l(y_i, \hat{y}_i)$ = loss function (e.g., squared error)
- $\Omega(f_k)$ = regularization term controlling model complexity
- n = number of training samples

The hyperparameters that are optimized during training include the learning rate, the depth of the tree, and estimators.

The trained models are then tested on the test data with regression measurements of R^2, MAE and RMSE.

H. Model Evaluation

Regression measures that are suitable to model performance are used to assess model performance. Evaluation measures such as the Coefficient of Determination of (R^2), Mean Absolute Error (MAE), as well as the Root Mean Square Error (RMSE) are considered key. These measures use the difference between measured cognitive scores and predicted ones with the ground truth to determine the accuracy of prediction. It is used in cross-validation in order to make certain that it is robust and to avoid overfitting.

1. Coefficient of Determination (R^2): The value of R^2 is used to understand the magnitude of the regression predictions which are close to the actual data values. It is the rate of variation of the dependent variable that can be given by the independent variables.

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \tag{6}$$

Where:

- y_i = actual value
- $\hat{y}_i$ = predicted value
- $\bar{y}$ = mean of actual values
- n = total number of samples

The closer value of R^2 to 1 it is, the better the model performs and the closer it is to 0; thus, predicting poorly.

2. Mean Absolute Error (MAE): MAE quantifies the entire value of the prediction errors regardless of their direction. It gives an easily interpretable prediction accuracy measure.

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \tag{7}$$

Where:

- y_i = actual value
- $\hat{y}_i$ = predicted value
- n = number of observations

A smaller value of MAE implies increased model accuracy.

3. Root Mean Square Error (RMSE): RMSE is a measure that quantifies the square root of the mean squared differences between the forecasted and the actual. It is more punitive of large errors than of MAE.

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{8}$$

Where:

- y_i = actual value
- $\hat{y}_i$ = predicted value
- n = total number of samples

A smaller value of RMSE shows performance results of a model and reduction of large prediction errors.

TABLE I
PERFORMANCE COMPARISON OF MRI-ONLY, BLOOD-ONLY, AND COMBINED MODELS

Model	R^2 Score	MAE	RMSE
MRI-only Model	0.2163	1.5606	2.1646
Blood-only Model	0.1075	1.7761	2.3098
Combined Model	0.2641	1.5946	2.0975

TABLE II
TRAINING AND TESTING R^2 PERFORMANCE

Model	Training R^2	Testing R^2
MRI-only Model	0.9788	0.2163
Blood-only Model	0.8087	0.1075
Combined Model	0.9629	0.2641

4. Cross-Validation: To make sure that the model is reliable and it does not overfit, k-fold cross-validation is used. Model In this method, the data is split into 8 folds numbering k. The model is trained using $k - 1$ folds and tested on the remaining fold. This is done again and again with k folds and each fold acts as a validation set. Average results in the evaluation metrics in each fold are averaged to obtain the final performance score.

TABLE III
CROSS-VALIDATION PERFORMANCE (MEAN R^2)

Model	Mean R^2 Score
MRI-only Model	0.1988
Blood-only Model	0.0810
Combined Model	0.2775

I. Model Interpretability Using SHAP

Explainable artificial intelligence strategies are included to provide clinical transparency with the help of SHAP (SHapley Additive Explainers). SHAP continues the ideas of cooperative game theory and describes the output of machine learning models as a result of giving every single feature an importance score based on its contribution to a given prediction. SHAP analysis measures the value that each feature has contributed to the final prediction, which allows determining which of the MRI features and blood biomarkers are most influential. The interpretation step enables the clinicians to determine the various potential anatomical or biochemical factors that have the strongest impact on the cognitive severity estimation.

SHAP Value Calculation: SHAP value gives the role that a feature plays in the prediction of a model by taking into consideration the number of combinations of features.

$$\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!}\left[f(S \cup \{i\}) - f(S)\right] \tag{9}$$

Where:

- ϕ_i = SHAP value representing the contribution of feature i
- F = set of all input features

- S = subset of features excluding feature i
- $f(S)$ = model prediction using feature subset S
- $f(S \cup \{i\})$ = model prediction when feature i is added to subset S

The SHAP framework makes sure that the total contribution of all the features is equal to the difference between the model prediction and the average prediction:

$$f(x) = \phi_0 + \sum_{i=1}^{M} \phi_i \tag{10}$$

Where:

- $f(x)$ = model prediction for input x
- ϕ_0 = base value (average model prediction)
- ϕ_i = contribution of feature i
- M = total number of features

SHAP-based interpretation enables clinicians to understand which of the MRI based structural measurements, as well as blood biomarker levels, are having the largest contribution to the consequence measure of cognitive severity, which enhances the model transparency and reliability of clinical decision support systems.

TABLE IV
MODALITY CONTRIBUTION BASED ON SHAP ANALYSIS

Modality	Contribution Score	Percentage Contribution
MRI Features	2.7213	72.28%
Blood Biomarkers	1.0437	27.72%

TABLE V
SHAP OUTPUT STATISTICS

Metric	Value
SHAP Value Matrix Shape	(100, 46)
Total Features Evaluated	46
Samples Used for Explanation	100

J. Local System Implementation

Owing to the local vision, the entire multimodal framework is deployed and operated to a local computer without involving the outer APIs and cloud infrastructure. The design will ensure privacy of data, reduce the cost of implementation and practicability can be deployed in a research environment or in a clinical environment. The system processes an input data by making interpretable results and executing model prediction internally in accordance with a self-governing localized environment.

V. SYSTEM ARCHITECTURE

The suggested system architecture will provide an approximation of cognitive severity in Alzheimer disease by combining brain features as extracted by structural MRI with peripheral blood biomarkers in a comprehensible system of multiple modalities of machine learning. The architecture is in a series of a pipeline consisting of data gathering, preprocessing, feature extraction, multimodal combination, training of the regression model, testing and interpretation of performance. The entire architecture operates on a local basis and does not rely on external application programmer interface or cloud execution calculation infrastructure, hence promoting data confidentiality, reproducibility, and advantageous experiment execution.

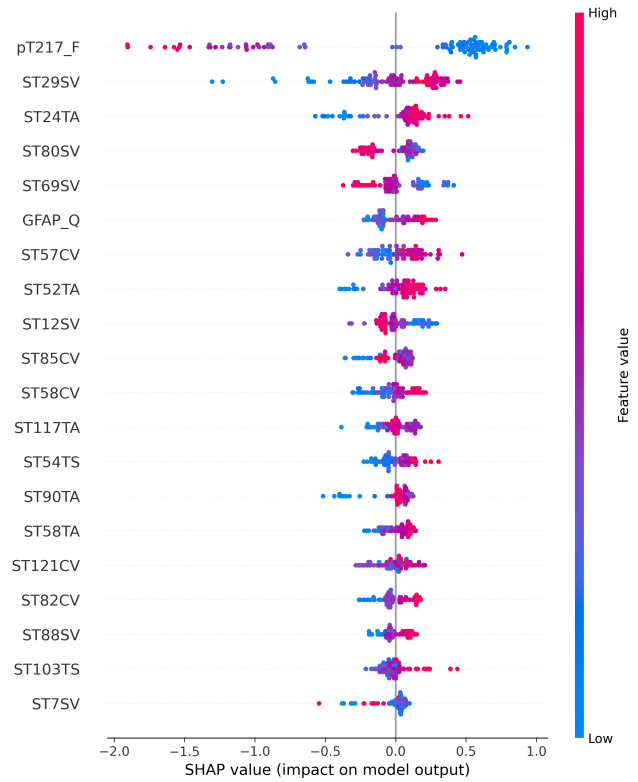

Fig. 2. SHAP Summary

The two dimensional modalities that can be fed to the system include structural features obtained by means of MRI as well as blood biomarker results. The variables obtained in MRI represent neuroanatomical changes related to disease progression such as the decrease of hippocampal volume, cortex thinning, and local region-specific changes in structure [Ref]. Blood biomarkers are are supplementary molecular data in the cases of amyloid-beta imbalance, tau dysregulation, and neuroinflammatory reaction [Ref]. The combination of these modalities allows the simultaneity of structural and biochemical processes of cognitive decline.

Any obtained information is handled by a preprocessing unit that is meant to enhance data integrity, and analytical accuracy. This step will entail the handle of missing values, feature normalization, outlier reduction, and feature cross-modal scaling. Then, feature selection and dimensionality reduction is used to eliminate redundant variables, add noise and enhance the computational efficiency, whilst maintaining the predictive information.

After feature refinement, a multimodal fusion is carried out by simply concatenating biomarker-derived features and MRI-

derived features into one:

$$X^{Multi} = [X^{MRI}, X^{Bio}] \tag{11}$$

Where:

- X^{MRI} = MRI-derived feature vector
- X^{Bio} = blood biomarker feature vector
- X^{Multi} = fused multimodal feature vector

This combined feature space allows the learning algorithm to learn to form interactions between anatomical and biochemical signals that such a methodology may fail to detect when data treated in isolation is analyzed.

A combination of this data is then trained to predict trends using models based on regression based ensemble learning, such as XGBoost and Random Forest, which was chosen due to its strengths in capturing nonlinear feature interactions and classes of heterogeneous data distributions. These models predict continuous scores of cognitive severity by training predictive relationships from multimodal input features to variables of clinical outcome.

The interpretability of the final stage is provided using SHAP analysis to enhance transparency of the model. SHAP calculates the value of individual input features to the predicted outcome based on:

$$f(x) = \phi_0 + \sum_{i=1}^{M} \phi_i \tag{12}$$

Where:

- $f(x)$ = model prediction
- ϕ_0 = baseline prediction
- ϕ_i = contribution of feature i
- M = total number of features

The interpretability layer allows one to identify the structural brain variables and blood biomarkers that have the largest impact on the estimation of cognitive severity.

The overall system output will be the estimated scores of cognitive severity with feature-level interpretability metrics on the relative contribution of each modality. This is a design of architecture that will trade off predictive performance with interpretability to facilitate both research-based analysis and the creation of future clinical decisions support system.

VI. Results and Discussion

The performance metrics that were used to evaluate the proposed multimodal framework were regression-based that are suitable in the approximate estimation of cognitive severity that is continuous. The aim of analysis was to evaluate predictive accuracy, and to find out whether predicted results of integration of the features of MRI with blood biomarkers are better when compared with one-modality representations.

A. Performance Evaluation Metrics

Quantitative evaluation was conducted using three measures of regression:

Coefficient of Determination (R^2): This measure is used in determining the degree of variation might in observable scores of cognitive severity that are dictated by model predictions. When values are higher it implies better predictive fit.

Mean Absolute Error (MAE): MAE is used to identify the average severity difference between predicted and observed scores, and it gives an estimate of the error of prediction which can be easily interpreted.

Root Mean Square Error (RMSE): RMSE measures the square root of the mean squared errors of prediction and resorts to high weights of the greater ones.

These measures are used to obtain complementary measures of predictive performance, model fit and measures of the magnitude of error.

B. Multimodal Performance Analysis

The results of the performed experiments provide evidence that integration of MRI generated structural features and blood biomarkers measurements results in better predictive performance compared to configurations of unimodal features. Neuroanatomical degeneration manifested in structural MRI variables and the biochemical processes complemented by blood biomarkers are the two ends of the cognition decline and disease pathology, respectively. Their joint representation allows the model to consider interactions in anatomical and molecular scale, which relate to the disease.

C. Feature Importance and Interpretability

SHAP-based model interpretability analysis found the most relevant predictors of cognitive severity estimation. Characteristics like subtraction of hippocampal volume and defective concentration of biomarkers were of significant impacts on predicting outputs that are congruent with the known patterns of neurodegeneration process reported in earlier experiments [Ref].

Interpretability framework enhances the results of transparency by measuring feature-level inputs, in this way facilitating their interpretation of model behavior in a way that may be considered to hold clinical significance.

D. Discussion of Findings

The suggested multimodal model proves that regression-based estimation of cognitive severity by combining structural neuroimaging characteristics and biochemical evidence can be enhanced. The way explainable artificial intelligence methods can be integrated into the framework also makes it stronger because it allows interpreting predictive drivers in an open way.

Despite promising performance of the current framework in controlled conditions of experimental settings, external validation should be done on larger and clinically diverse data sets prior to translational implementation. Future studies should take into account increased samples of cohorts, other categories of biomarkers and verification in practical clinical conditions.

VII. Conclusion

This paper provides a multimodal machine learning system to predict the cognitive degradation in Alzheimer disease by integrating structural MRI obtained features with external peripheral blood biomarker values. Integrating neuroanatomical and biochemical measures in a single regression-based learning model, the proposed solution will enhance the prediction of continuous severity of cognitive scores and retain the interpretability of the model.

The experimental findings suggest that multimodal integration of features offers a better predictive ability in contrast to single-modality of analysis due to the handling of complementary information related to the disease progression. The nonlinear interactions between the imaging and biomarker variables were successfully modeled with ensemble regression models, such as the XGBoost and Random Forest, and with SHAP-based interpretations, it was also possible to identify influential predictors that led to the estimation of cognitive severity.

It provides the effect of transparency, where explainable artificial intelligence can help in a transplant of feature-level explanations, which are still clinically interpretable and distinctly correlated with familiar disease-related structural and biochemical marks [Ref]. This contributes to the possible usefulness of the framework in cognitive severity research-based assessment and decision-support development in the future.

However, regardless of such encouraging results, there are a number of limitations. The framework must also be tested more extensively on generalizability on clinical populations by using larger and more heterogeneous datasets. It can also be suggested that in the future, other imaging forms, longitudinal measurements of biomarkers, and external validation in the clinical environment can be introduced to strengthen and increase its translational usefulness.

In general, the suggested framework shows that a multimodal integration with an interpretable machine learning can provide a good way to enhance computational evaluation of the cognitive degradation in Alzheimer disease.

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